

SWORD v17c Beta Release Notes

Version: v17c beta 0.0.4 **Date:** April 2026 **Authors:** James H. Gearon, Tamlin M. Pavelsky, Niek Collot d'Escury **Base version:** SWORD v17b (March 2025, UNC)

Changelog

0.0.4 (March 2026)

- **Node propagation to 11.1M nodes.** Five v17c columns (`best_headwater`, `best_outlet`, `pathlen_hw`, `pathlen_out`, `subnetwork_id`) were NULL on all nodes. Now propagated from parent reaches.
- **Dijkstra ghost outlet fix.** Ghost reaches (`type=6`) with `out_degree=0` no longer report `dist_out_dijkstra=0`. All sinks are used as Dijkstra sources for full coverage (94–99% per region); ghost sinks receive NULL. Real outlet counts: NA=7, SA=1, EU=2, AF=1, AS=11, OC=1.
- **Bifurcation routing fix.** At multi-successor nodes, `rch_id_dn_main` now follows the mainstem chain unconditionally (was falling back to score-based ranking, which could disagree with `is_mainstem`). V023 (`pathlen_out` step consistency) violations: $2 \rightarrow 0$.
- **hydro_dist_hw computed.** Mainstem distance from headwater via `rch_id_up_main` chain walk (mirror of `hydro_dist_out`). Was stale from a prior pipeline run.
- **facc monotonicity fix (T003).** 419 reaches corrected via iterative downstream propagation. T003 violations: $392 \rightarrow 0$.
- **Routing weights retrained on effective_slope.** SWOT `slope_obs_p50` (where reliable and $n \geq 5$) replaces MERIT DEM slope in routing score training. Slope share: $8\% \rightarrow 3\%$, width: $67\% \rightarrow 71\%$. CV accuracy unchanged (88.5%).
- **Node-level distance interpolation.** `hydro_dist_out`, `hydro_dist_hw`, and `dist_out_dijkstra` added to nodes table, interpolated by node position within reach (using v17b `dist_out` offset). `pathlen_hw` and `pathlen_out` changed from flat reach copies to per-node interpolated values.
- **hydro_dist_hw convention fix.** Reach-level values shifted from downstream-end anchor to upstream-end anchor, matching `dist_out`, `hydro_dist_out`, and `dist_out_dijkstra`. Headwater reaches now report 0 (was `reach_length`). Node-level values unchanged.
- **F006 junction conservation fix.** 2 remaining junction violations (OC 53130100215, AS 45311901585) resolved by setting downstream `facc` to sum of upstream `facc`. F006 violations: $2 \rightarrow 0$.
- **13 AS main_side reverted to v17b.** Reaches had `main_side` changed from 0 (main) to 1 (side) by an undetermined prior operation; 9 of 13 are linear reaches where side-channel classification is impossible.
- **Distance convention documented.** Variable reference now includes a convention table specifying the measurement anchor, zero-point, and ghost reach behavior for all distance variables, plus node-level interpolation formulas.
- **Variable reference updated.** 7 missing variables added, 8 type mismatches fixed, `cl_ids` shape corrected to `cl_id_min/cl_id_max`.
- **Node dist_out reactive recalc fix.** `CALCULATE_DIST_OUT` operations now correctly recalculate node-level `dist_out` by sorting nodes via `node_order` (geometric position) instead of `node_id`. This fixes an issue where flow-corrected reaches had incorrect node distances after recalculation (e.g., downstream-most node getting downstream reach's `dist_out`).
- **Ghost coastal outlet dist_out_dijkstra fix.** Isolated ghost coastal outlets (`type=6` sinks with upstream neighbors but no path to real hydrologic outlets) now receive `dist_out_dijkstra = reach_length`, matching v17b `dist_out` behavior. Previously these 703 reaches in NA (similar counts in other regions) had NULL, but they should report the distance from their start to the ocean outlet point.
- **Exports regenerated (April 1, 2026).** All NetCDF, GeoPackage, and Parquet files updated with the above fixes. SHA256SUMS refreshed.

0.0.3 (March 2026)

- **Routing weights learned from human labels.** Replaced the handcrafted lexicographic 3-tuple (`effective_width`, `log_facc`, `pathlen`) with a weighted scalar score trained on 1,967 human-labeled junction decisions: $1.97\log_{10}(ew) + 0.23\log_{10}(facc) - 0.23\log_{10}(slope) + 0.23\log_{10}(pathlen) - 0.29*stream_order$. Two new signals vs prior releases: `slope` (negative = prefer lower gradient) and `stream_order`. All routing functions use the same score to prevent divergence. (Weights retrained in 0.0.4; see above.)
- **Caroline’s reviewer fixes synced.** 127 C001 lakeflag fixes (NA) and 10 C004 type fixes (NA) from the Streamlit reviewer app.
- **NA PostgreSQL geometry fix.** 38,696 NA reaches had NULL geometry in PostgreSQL — v17b geometry overwrite via dblink silently failed. Rewrote to read from v17b GPKG files on disk with verification gates.
- Fixed `dn_node_id`, `up_node_id`, and `node_order` for 810 flow-corrected reaches where node ordering was stale (based on v17b `dist_out`). 639 reaches now have `dn_node_id != min(node_id)`, reflecting the inverted flow direction.
- Removed redundant `rch_id_up_1..4` / `rch_id_dn_1..4` vector variables from NetCDF export. Only the [4, N] matrices `rch_id_up` and `rch_id_dn` are exported, matching v17b format.
- Fixed `swot_orbits` NetCDF type from string back to int64 (matching v17b).
- Fixed 15 integer columns widened from int32 to int64 in NetCDF export (`n_nodes`, `lakeflag`, `n_rch_up`, `stream_order`, etc.). All shared variable types now match v17b exactly.
- Synced `rch_id_up_1..4` / `rch_id_dn_1..4` DB columns from `reach_topology` table (694 were stale after flow corrections).

0.0.2 (March 2026)

- Renamed `is_mainstem_edge` to `is_mainstem`
- Mainstem algorithm refactored: computed per `main_path_id` group (see Section 2.1)
- Fixed `n_rch_up`/`n_rch_down` stale counts at 148 flow-corrected reaches
- Recomputed `facc` at 807 flow-corrected reaches using topological propagation; resolves all F006 junction conservation violations and T003 monotonicity violations at these reaches
- OC reach 51111300061 incomplete split fully reverted to v17b state (434 orphan centerlines, 73 orphan nodes restored)

0.0.1 (March 2026)

- Initial v17c beta release
- Includes `node_order`, `dn_node_id`, `up_node_id` (node ordering within reaches)

1. Overview

SWORD v17c extends v17b with three additions: computed mainstem topology, SWOT observation statistics, and flow accumulation corrections. No reaches, nodes, or centerlines were added or removed. v17c contains the same 248,673 reaches, 11.1M nodes, and 66.9M centerline points as v17b across all six regions (NA, SA, EU, AF, AS, OC).

Each region is distributed as a single NetCDF4 file (`{region}_sword_v17c_beta.nc`). The group structure matches v17b (centerlines, nodes, reaches), and the `area_fits` and `discharge_models` subgroups under reaches pass through from v17b unchanged. Reach and node ordering within each file matches v17b.

Reach coordinate columns (`x`, `y`, `x_min`, `x_max`, `y_min`, `y_max`) match v17b values across all formats (NetCDF, DuckDB, PostgreSQL).

All new variables use a fill value of -9999 where no observation or computation produced a value.

For a complete variable catalog, see `v17c_variable_reference.md`.

2. New Variables

2.1 Mainstem Topology (reaches group)

`is_mainstem` is computed per `main_path_id` group: each group (a mainstem path plus its tributary branches) gets one canonical chain, identified by a greedy walk from the group's shared `best_headwater`. At each junction the algorithm selects the upstream branch with the highest weighted routing score: $2.02 \cdot \log_{10}(\text{ew}) + 0.17 \cdot \log_{10}(\text{facc}) - 0.08 \cdot \log_{10}(\text{slope}) + 0.35 \cdot \log_{10}(\text{pathlen}) + 0.23 \cdot \text{stream_order}$. These weights were learned from 1,967 human-labeled junction decisions via logistic regression on pairwise $\log_{10}$ -difference features, using SWOT slope where reliable (else MERIT DEM). The negative slope weight captures the geomorphic pattern that mainstem channels have lower gradients than tributaries. Mainstem reaches within each group are then assigned `rch_id_up_main` / `rch_id_dn_main` from the chain; non-mainstem reaches use the same weighted score. Ghost reaches (type=6) are excluded from mainstem but still participate in routing topology. ~10% of mainstem reaches have no mainstem neighbor at `main_path_id` group boundaries — this is expected by design.

Variable	Type	Units	Description
<code>dist_out_dijkstra</code>	float64	meters	Dijkstra shortest-path distance from the upstream end of the reach to any network outlet (outlet = 0; NULL for ghost reaches)
<code>hydro_dist_out</code>	float64	meters	Mainstem distance from the upstream end of the reach to <code>best_outlet</code> via <code>rch_id_dn_main</code> chain (outlet = <code>reach_length</code>)
<code>hydro_dist_hw</code>	float64	meters	Mainstem distance from <code>best_headwater</code> to the upstream end of the reach via <code>rch_id_up_main</code> chain (headwater = 0)
<code>rch_id_up_main</code>	int64	—	Main upstream neighbor reach_id (mainstem-preferred)
<code>rch_id_dn_main</code>	int64	—	Main downstream neighbor reach_id (mainstem-preferred)
<code>best_headwater</code>	int64	—	Routing-score-prioritized headwater reach_id for the network component
<code>best_outlet</code>	int64	—	Routing-score-prioritized outlet reach_id for the network component
<code>pathlen_hw</code>	float64	meters	Cumulative path length from <code>best_headwater</code>
<code>pathlen_out</code>	float64	meters	Cumulative path length to <code>best_outlet</code>
<code>is_mainstem</code>	int32	—	1 if reach is on a mainstem path, 0 otherwise
<code>main_path_id</code>	int64	—	Unique identifier for each mainstem path group
<code>subnetwork_id</code>	int32	—	Connected component ID (Pfaffstetter-offset, globally unique; see Section 4)

Variable	Type	Units	Description
dn_node_id	int64	—	Node ID at the downstream end of the reach (lowest <code>dist_out</code>)
up_node_id	int64	—	Node ID at the upstream end of the reach (highest <code>dist_out</code>)

Eight of these variables also appear at node level. Five are interpolated by node position within the reach: `hydro_dist_out`, `hydro_dist_hw`, `dist_out_dijkstra`, `pathlen_hw`, and `pathlen_out`. Three are flat copies from the parent reach: `subnetwork_id`, `best_headwater`, and `best_outlet`. For flow-corrected reaches (660), the interpolation reverses the offset direction to produce correct monotonic distances.

The interpolation offset for each node is `reach.dist_out - node.dist_out`, where `node.dist_out` is the v17b original (not recomputed). This offset equals 0 at the upstream end of the reach and `reach_length` at the downstream end. For single-node reaches the offset is `reach_length / 2` (midpoint). Node-level `dist_out` itself is the v17b value, preserved as-is for backward compatibility; at single-node reaches it equals the reach-level value rather than the midpoint.

At reach boundaries the downstream-end node of the upstream reach and the upstream-end node of the downstream reach share identical interpolated distance values (verified gap = 0 m across all boundaries).

`node_order` is a node-level variable (not in the reaches table): 1-based position within a reach, ordered by `dist_out` ascending (1 = downstream end, n = upstream end).

Distance convention note. All four distance variables anchor at the upstream end of the reach. `dist_out` (v17b) and `hydro_dist_out` assign `reach_length` at the outlet. `dist_out_dijkstra` assigns 0 at the outlet; the offset at any reach equals the outlet's `reach_length`. `hydro_dist_hw` assigns 0 at the headwater and increases downstream. See the variable reference for the full convention table.

2.2 SWOT Observation Statistics

Percentile-based summaries computed from available SWOT observations. All percentile, range, and MAD variables share the units of the underlying measurement.

Reaches and nodes:

Variable	Type	Units	Description
wse_obs_p10– wse_obs_p90	float64	meters	WSE percentiles (10th through 90th, in steps of 10)
wse_obs_range	float64	meters	WSE observation range (p90 - p10)
wse_obs_mad	float64	meters	WSE median absolute deviation
width_obs_p10– width_obs_p90	float64	meters	Width percentiles
width_obs_range	float64	meters	Width observation range
width_obs_mad	float64	meters	Width median absolute deviation
n_obs	int32	—	Total SWOT observation count

Reaches only:

Variable	Type	Units	Description
slope_obs_p10– slope_obs_p90	float64	m/km	Slope percentiles
slope_obs_range	float64	m/km	Slope observation range
slope_obs_mad	float64	m/km	Slope median absolute deviation

Variable	Type	Units	Description
slope_obs_adj	float64	m/km	Adjusted slope
slope_obs_slopeF	float64	—	Slope F-statistic
slope_obs_reliable	int32	—	0 = unreliable, 1 = reliable
slope_obs_quality	int32	—	Integer quality category (0–8; see Section 3)
slope_obs_n	int32	—	Number of slope observations
slope_obs_n_passes	int32	—	Number of SWOT passes used
slope_obs_q	int32	—	Bitfield quality flag (see Section 3)

2.3 Flow Accumulation Corrections

A two-stage denoise pipeline corrected flow accumulation (**facc**) values to address three systematic error modes in MERIT Hydro’s D8 (eight-direction flow routing) upstream area: bifurcation cloning, junction inflation, and raster-vector misalignment. The pipeline corrected 96,589 of 248,673 reaches (38.8%). Uncorrected reaches retain v17b values. See `facc_correction_methodology.md` for the full algorithm description.

Variable	Type	Group	Description
facc	float64	reaches, nodes	Flow accumulation (km ²). Corrected values where applicable; v17b values otherwise.
facc_quality	int32	reaches, nodes	1 = corrected by denoise_v3; fill_value = not flagged

After correction, junction conservation violations (downstream **facc** < sum of upstream **facc**) are resolved in all regions. In 0.0.2, **facc** was additionally recomputed at 807 flow-corrected reaches via topological propagation, resolving remaining monotonicity violations at those reaches.

2.4 Other New or Updated Variables

Variable	Type	Group	Description
type	int32	reaches	Reach classification (1=river, 3=lake_on_river, 4=dam, 5=unreliable, 6=ghost). Not present in v17b NetCDF; added in v17c so NetCDF users can filter by reach type without needing the database.
dl_grod_id	int64	reaches	DL-GROD (Deep Learning Global River Obstruction Database; He et al. 2025) dam/obstruction ID
edit_flag	string	reaches	Tag for manually edited reaches (e.g., <code>lake_sandwich</code> , <code>harp_lake</code>)

3. Flag Encoding Reference

facc_quality

Value	Meaning
1	denoise_v3 — corrected by the facc denoising pipeline
-9999 (fill)	Not flagged; facc unchanged from v17b

CF attributes: `flag_values = [1]`, `flag_meanings = "denoise_v3"`.

slope_obs_quality

Value	Meaning
0	reliable
1	small_negative
2	moderate_negative
3	large_negative
4	negative
5	below_ref_uncertainty
6	high_uncertainty
7	noise_high_nobs
8	flat_water_noise

CF attributes: `flag_values = [0,1,2,3,4,5,6,7,8]`, `flag_meanings = "reliable small_negative moderate_negative large_negative negative below_ref_uncertainty high_uncertainty noise_high_nobs flat_water_noise"`.

slope_obs_reliable

Value	Meaning
0	Unreliable
1	Reliable
-9999 (fill)	Not computed (no SWOT slope observations)

slope_obs_q (bitfield)

Bit	Value	Meaning
1	1	Negative slope
2	2	Low number of passes
3	4	High variance
4	8	Extreme value
5	16	Clipped

Flags combine by addition. A value of 0 indicates no quality issues. Example: 5 = negative slope (1) + high variance (4).

is_mainstem

Value	Meaning
0	Not on mainstem

Value	Meaning
1	On mainstem path

4. Known Limitations

- **SWOT observation coverage:** SWOT statistics are `fill_value` (-9999) for reaches and nodes lacking SWOT data.
- **facc correction scope:** 96,589 reaches corrected (38.8%); the remaining 152,084 retain v17b values. Node-level facc propagates from the parent reach.
- **Lake sandwich corrections:** 1,252 reaches reclassified to `lakeflag` = 1 where a narrow, shorter-than-neighbor reach sat between lake reaches (tagged `edit_flag` = "lake_sandwich"). ~1,755 similar cases remain (narrow connecting channels, chains).
- **HarP lake corrections:** 7,425 reaches reclassified from `lakeflag` = 0 (river) to `lakeflag` = 1 (lake) based on HarP v1.1 (Hydrography and River Planform) lake classification data. 200,201 child nodes updated to match. Tagged `edit_flag` = "harp_lake". Existing tags preserved (comma-delimited when multiple apply).
- **area_fits and discharge_models:** Direct copies from v17b. Not recomputed against v17c facc or SWOT values.
- **subnetwork_id vs network:** `subnetwork_id` uses Pfafstetter- offset enumeration (globally unique). v17b `network` uses per-region 1-based IDs. Different component counts (v17c finds more via weakly connected components; 19 subnetworks span multiple v17b networks). `network` is retained unchanged from v17b.
- **Topology reciprocity gaps (resolved):** 151 non-reciprocal pairs (both reaches listing each other as upstream) were introduced during flow correction revert. Fixed by completing the revert for all affected reciprocal entries. Zero non-reciprocal pairs remain.
- **Flow correction oscillation:** 389 reaches (0.16%) in AF/AS/EU/NA/SA had ambiguous WSE slope signals causing bidirectional flow correction scores. These were reverted to v17b topology.
- **main_path_id consistency:** 3,134 reaches have `main_path_id` values inconsistent with current (`best_headwater`, `best_outlet`) tuples (V013- V015 lint checks). 80 reaches in NA have `best_headwater` pointing to non-headwater reaches. Requires recomputing `main_path_id` from current headwater/outlet assignments.
- **River naming:** 51.2% of reaches are unnamed (NODATA), ranging from 26% (AF) to 69% (OC). 2.6% of mainstem 1:1 links have local name discontinuities (name changes between adjacent reaches with no junction).
- **Width fill values (A003):** ~1,266 reaches have `width`=0 (unmeasured) or `width`=-1 (GRWL lake fill). These are v17b fill values, not data errors. Present in both v17b and v17c unchanged. The A003 lint check is downgraded to WARNING for this reason.
- **lakeflag/type mismatch:** ~5,770 reaches have `lakeflag`=1 (lake) but `type`=1 (river), introduced by HarP and lake-sandwich corrections which updated `lakeflag` but not `type`. `type` is encoded in the last digit of `reach_id`, so changing it would change reach IDs. Policy for `reach_id` changes under discussion. Affects users filtering on both `lakeflag` and `type` at river-lake boundaries. Tracked in issue #208.

5. Quality Audits

Validation checks performed on the v17c data:

Audit	Finding
Geometry	DuckDB geometries (rebuilt from NetCDF) lack endpoint overlap vertices present in v17b (210,533 reaches affected: 173K +1 point, 37K +2 points). <code>reach_length</code> unchanged. Reach coordinate columns (<code>x</code> , <code>y</code> , <code>x_min</code> , <code>x_max</code> , <code>y_min</code> , <code>y_max</code>) copied from v17b to ensure consistency across all formats.
n_nodes / reach_length path_freq gaps	Internally consistent. Zero N008/G002/G003 violations. v17b had 4,952 connected non-ghost reaches with invalid <code>path_freq</code> (0 or -9999). Resolved in v17c; remaining nodata values are correctly attributed to ghost reaches (type=6).
subnetwork_id	3,027 components across 248,673 reaches verified. Pfafstetter banding correct. Zero cross-region collisions. 19 subnetworks (0.6%) span multiple v17b networks (expected).
Topology integrity	T001 (<code>dist_out_dijkstra</code> monotonicity), T012 (referential integrity), T013 (self-reference), T014 (bidirectional): all pass. T005/T007: zero non-reciprocal edges (151 from incomplete flow correction revert resolved in beta 0.0.1). Note: v17b <code>dist_out</code> is stale at 807 flow-corrected reaches where topology direction was updated but <code>dist_out</code> retains its v17b value — use <code>dist_out_dijkstra</code> or <code>hydro_dist_out</code> for distance routing.
n_rch_up/n_rch_down	148 scalar count mismatches corrected (flow corrections flipped <code>reach_topology</code> but did not recalculate counts). Zero mismatches across all 248,673 reaches.
OC reach split revert	Incomplete <code>break_reaches()</code> split of OC reach 51111300061 (434 orphan centerlines, 73 orphan nodes) fully reverted to v17b state.
River name formatting	291 formatting issues corrected (separators, whitespace). Automated checks now enforce “;” separator and alphabetical ordering.
Flow direction	1,112 experimental topology flips reverted after causing 30K disconnected reaches. 807 reaches across all regions retain corrected topology that differs from v17b (175 sections, median 5 reaches/section). OC has 119 of these (26 SWOT-validated sections). Non-OC corrections were retained from the flow correction pipeline; <code>dist_out</code> is stale at these reaches — use <code>dist_out_dijkstra</code> .
HarP lake corrections	7,425 reaches reclassified lakeflag 0 to 1 from HarP v1.1 data. 200,201 nodes propagated. Tagged <code>edit_flag</code> = “harp_lake”.

For POM (Pierre-Olivier Malaterre) validation results, see `pom_validation_report.md`.

6. File Format

- **Format:** NetCDF4 (one file per region)
 - **Naming:** {region}_sword_v17c_beta.nc where region is na, sa, eu, af, as, oc
 - **Groups:** centerlines, nodes, reaches
 - reaches/area_fits and reaches/discharge_models subgroups (from v17b)
 - **Ordering:** Reach, node, and centerline arrays match v17b ordering
 - **Fill value:** -9999 for all numeric variables (int32, int64, float64)
 - **Checksums:** SHA256 hashes listed in SHA256SUMS_{version}.txt
 - **Additional formats:** GeoPackage and GeoParquet exports available (reaches and nodes per region, with geometry)
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7. Methodology Documentation

Document	Description
facc_correction_methodology.md	Facc denoise algorithm, detection rules, correction model
pom_requests_summary.md	POM validation check tracker (19 checks, production results)
pom_validation_report.md	POM validation production results
v17c_variable_reference.md	Complete variable catalog for NetCDF export
SWORD_v17b_Technical_Documentation.md	v17b baseline reference